A PhD student tells her supervisor she's finished five chapters. The supervisor reads them and says, "Congratulations. You've also started five theses."

Earn the Right to Ask

Earning your Gap

A gap is an argument, not an absence. This argument has to run through every chapter.

Table, Track, Terms….

YOUR TABLE

Humanities · Social Sciences · STEM · Math · Design and Architecture

Sit by how your thesis makes its argument, not by its topic.

Fewer than three at a table? Merge. More than six? Split.

YOUR TRACK

Read. No topic yet. Work on a published thesis.

Plan. A topic, gap still unclear.

Repair. You are drafting.

THREE TERMS

The gap. An argument for why a question needs answering.

The key noun. The one word your gap is about.

The thread. The gap carried through every chapter.

PART I OF III

Earn the Gap

Is my gap an argument, or only an absence?

Why an absence is not a gap · The three tests · The four cracks · Gap or Trap · The gap picks the method

I · EARN THE GAP

II · CARRY THE GAP

III · LINK THE REVIEW

Defend This Sentence

> *"A gap is simply something nobody has studied."*

Two minutes at your table. Make the best case you can for it.

Most of you believe some version of this. That is why we start here.

An Absence is not a Gap

Nobody has studied the lunch habits of left-handed ferry captains.

The literature is silent.
The silence is deserved.

An absence is a fact.

A gap is an argument.

Your thesis is not just a gap finding-solving journey. It builds the case for one.

Test the Gap: Turning An *Absence* Into A *Gap*

1

Stakes not mere Absence: The consequence test

What goes wrong, in theory or in practice, if this stays unanswered? If nothing goes wrong, you have a hole, not a gap.

2

Answerable, not merely interesting: The adjudication test

Could evidence settle it? In conceptual fields, argument and cases help settle it. In mathematics, proof or counterexample. If nothing could, it is a seminar, not a thesis.

3

Verified, not borrowed: The inheritance test

Who takes up your answer once you have it? Name the reader downstream. If no one inherits it, the field will not miss it.

Three Tests Turn An *Absence* Into A *Gap*

1

The consequence test proves the gap actually matters

Fails: "No study has examined retention among rural dental practitioners in New Brunswick."
Passes: "Provincial projections assume rural dentists leave at urban rates. If they leave faster, the 2030 model under-funds rural training seats."
Why it passes: It locates an untested assumption inside a model already making decisions.

2

The adjudication test proves the gap can actually close

Fails: "It is unclear whether high-temperature fatigue models apply to this alloy."
Passes: "Fatigue tests at 310°C and 340°C, on either side of the 320°C phase change, will show whether the 300°C data still predicts failure."
Why it passes: A bounded experiment can confirm or refute the extrapolation.

3

The inheritance test proves the gap still exists

Fails: "No one has studied the marginalia in the library's copy of the poem"
Passes: "If the political reading predates its 1840s editors, reception historians lose their origin story, and the forthcoming critical edition must re-date the reading."
Why it passes: It names who must revise their work.

The Four Cracks Show Where To Look for a Gap

1 Tension

Two credible bodies of work cannot both be right.

Synthesis tools average toward consensus. “Taken together” is the sound of a fight being buried.

2 Absence

Nothing there, and the three tests say there should be.

“No study has examined…” is a claim about the whole literature. Earn it.

3 Limit

True there. Untested here.

“Not yet done in Singapore” is geography. Name the mechanism that changes when the context does.

4 Assumption

The field agrees on something it has never tested.

Highest reward, highest risk. Early on, spot it in published work. Stake a thesis on it only with your supervisor signed on.

Every defensible research question starts in one of these.

The Four Cracks Show Where To Look for a Gap

1 **Tension**

Researchers disagree

1 **Assumption**

Researchers agree without having tested it

Both are about how much the existing Research agrees

2 **Absence**

No one has looked

2 **Limit**

Looked but only in one place

Both are about how much of the territory the existing research covers

Every defensible research question starts in one of these.

SHEETS OUT

Activity One

Gap or Trap

---

12 minutes, including the debrief.

Presenting stops here. You are on.

Gap or Trap

Five claims on your sheet. At your table, sort each into a bin.

Real gap

Passes all three tests.

Trivial absence

Unstudied because unimportant.

Filled gap

Studied. Your search was shallow.

Unanswerable

No evidence could settle it as posed.

Some claims survived. Which crack did it come from, and what study would it need?
The method follows from the crack.

Which Test Each Claim Failed

For returning to later. The full claims are on your activity sheet.

1	Whether democracy is the best form of government	**Unanswerable**	Fails adjudication as posed.
2	A stewardship protocol in clinics with no pharmacist	**Real gap**	A Limit crack with a named mechanism.
3	Whether human beings have free will	**Unanswerable**	Fails adjudication as posed.
4	Access to daylight & emotional impact on humans	**Trivial absence**	Fails consequence and inheritance.[Some might say filled gap]
5	The problems with a coloring bond	**Filled gap**	Decades of work. The search was shallow.

Which Test Did It Fail?

Each table reports one claim. Not the bin. The test.

Some claims survived. Which crack did it come from, and what study would it need?
The method follows from the crack.

The Gap Dictates The Tool

Tension	An adjudicating design. Built so the rival explanations make different predictions.
Absence	A mapping design. Exploratory reach,and claims modest enough to match.
Limit	Replication with variation. Change exactly the thing the original was bound to.
Assumption	A critical design. And a supervisor's signature before anything else.

If your method was chosen by department habit, the gap did not choose it.

PART II OF III

Carry the Gap

Does the same gap survive all five chapters?

Drift and the key noun · Five jobs, one per chapter · One whole thread · Thread or tape · Fix the Thread

DO · 3 MINUTES

Spot the Drift

Five sentences from one thesis, one per chapter.

INTRODUCTION	*"This thesis asks how small museums decide which objects to deaccession."*
LITERATURE REVIEW	*"Research on museum collection practices has largely overlooked smaller institutions."*
METHODS	*"Interviews with curators explore their collecting strategies."*
DISCUSSION	*"The findings show that governance shapes collection management."*
CONCLUSION	*"This study contributes to understanding museum sustainability."*

Where did this thesis stop being about one thing?

One Thesis, Three Gaps

CHAPTER 1	CHAPTER 2	CHAPTER 3	CHAPTER 4	CHAPTER 5
Decide	**Practices**	**Collecting**	**Management**	**Sustainability**

The key noun in each sentence of Spot the Drift. Each sentence is fine alone. Together they describe five different theses.

The key noun rule. Pick the one word your gap is about. Make every chapter carry it. When the noun changes, the claim has changed, whether you meant it to or not.

The GAP Does A Different Job In Every Chapter

	JOB	THE QUESTION IT ANSWERS
1	**Announce and stake**	What is missing, and who pays for it?
2	**Build the case**	Why can't existing work answer it?
3	**Make it answerable**	Why/How can this design see what earlier designs could not?
4	**Rule on it**	How far did the gap close?
5	**Account and hand on**	What can the field say now that it could not before?

Same gap, same key noun, five jobs. These are jobs, not chapter titles. Where each one sits depends on your field.

Announce & Stake

The introduction tells the reader what is missing, and who pays if it stays missing.

WEAK VERSION · TYPICAL OF AI DRAFTS

"Despite extensive research on antibiotic stewardship, its application in rural contexts remains underexplored."

STRONG VERSION · STEM EXAMPLE

"Stewardship protocols cut inappropriate prescribing in teaching lhospitals, where a pharmacist reviews every prescription. Rural clinics are adopting the protocol without that reviewer."

The move. Name the territory, the crack, the cost and the question. The cost sentence is the consequence test written as prose.

WHERE THIS JOB SITS IN YOUR FIELD

Humanities

Introduction.

Social Sciences

Introduction.

STEM

Introduction.

Math

Introduction, with the main theorem stated early.

Design

Introduction, framed as a problem in practice as well as in the literature.

Build the Case

The literature review shows why existing work cannot answer the question yet.

WEAK VERSION · TYPICAL OF AI DRAFTS

"While scholars have shed light on the novel's ending, further research is needed."

STRONG VERSION · HUMANITIES EXAMPLE

"Both camps read only the published ending. Neither consults the four surviving drafts, where the flood was written and rewritten."

The move. Organise the review around where each body of work stops. End each section with what that work cannot see.

WHERE THIS JOB SITS IN YOUR FIELD

Humanities

Spread across chapters as a critical conversation.

Social Sciences

Literature review.

STEM

Literature review.

Math

Background and known results, showing where existing proofs stop.

Design

Precedent review of built works alongside the literature.

Make it Answerable

The methods show why your design can see what earlier designs could not.

WEAK VERSION · TYPICAL OF AI DRAFTS

"A research-through-design approach was adopted, as it is well suited to exploring complex spatial phenomena."

STRONG VERSION · DESIGN EXAMPLE

"This thesis tests three versions of one sky garden, varying shade, distance from the lift lobby and seating. The prototypes can show which features draw use in that block. They cannot show whether use builds community."

The move. Name the blind spot of earlier designs. Choose a design that can see it. Then say what yours still cannot see.

WHERE THIS JOB SITS IN YOUR FIELD

Humanities

A declared reading position, corpus or lens.

Social Sciences

Research design.

STEM

Methods.

Math

Proof strategy, naming the step that fails and what replaces it.

Design

Prototypes or studio work as the instrument.

Rule On It

The discussion goes back to the gap and says how far your findings closed it.

WEAK VERSION · TYPICAL OF AI DRAFTS

"These findings fill the gap in the literature and contribute to a deeper understanding of intergroup contact."

STRONG VERSION · SOCIAL SCIENCES EXAMPLE

"Effects lasted only where contact was repeated and cooperative. The gap is reframed. The question is no longer whether contact works but what structure it needs."

The ruling scale. Closed, narrowed, reframed or reopened. Pick one and give the reason.

WHERE THIS JOB SITS IN YOUR FIELD

Humanities

End of each chapter, then the final chapter.

Social Sciences

Discussion.

STEM

Discussion.

Math

Remarks on sharpness and counterexamples.

Design

Evaluation or critical reflection chapter.

Account & Hand On

The conclusion states what the field can now say, and what it still cannot.

WEAK VERSION · TYPICAL OF AI DRAFTS

"Future research could explore this property in other classes of graphs."

STRONG VERSION · MATH EXAMPLE

"The field could say property P holds when no vertex is large. It can now say P holds when vertices are large on average only within limits. It still cannot say what happens between those limits and unbounded growth."

The triad. Could say. Can now say. Still cannot say.

WHERE THIS JOB SITS IN YOUR FIELD

Humanities

Coda or conclusion.

Social Sciences

Conclusion.

STEM

Conclusion.

Math

Open problems.

Design

Conclusion, separating contribution to knowledge from contribution to practice.

The Whole Thread, In One Thesis

One fictional business thesis, one sentence from each job. Watch the key noun.

1	ANNOUNCE	Innovation research tells firms how to launch. It says almost nothing about how they **decide** to stop.
2	BUILD	Neither literature has observed the **decision** itself. Both infer it from which products were still standing afterwards.
3	ANSWERABLE	The design can show how **decisions** unfold. It cannot estimate how common each pattern is.
4	RULE	This narrows the gap. Six firms in two industries show that retirement **decisions** are deferred, not how often.
5	ACCOUNT	It still cannot say whether deferred **decisions** dominate across industries.

Watch the handover

The limit stated in line 3 becomes the ruling in line 4.

The ruling becomes what the field still cannot say in line 5.

The key noun never changes.

Thread Or Tape?

Keeping the key noun does not mean repeating the same sentence.

A thread

The same claim, carried forward. Each chapter adds what it now knows.

Tape

The same sentence pasted into every chapter. The claim never moves.

The test. Does your Chapter 5 sentence know something your Chapter 1 sentence did not?

SHEETS OUT

Activity Two

The Thread Card

16 minutes.

Fix the Thread

Your card is one thesis in five lines, one per job. Most lines are broken. One or two are sound.

1

Find the breaks

Mark each line sound or broken.

2

Name the mistake

Choose from the list on your sheet.

3

Fix it

Rewrite each broken line. Keep the key noun.

THE MISTAKES TO LOOK FOR

No stake. Nobody pays if it stays missing.
Survey, not a case. Lists sources, never says where they stop.
Orphan. A topic no later line uses.
Inherited method. Copies the design that made the gap.
Wrong measure. Measures something other than the key noun.
No limit. Never says what the design cannot see.
Overclaim or no ruling. "Fills the gap", or no verdict at all.
Stranger. A new source suddenly explains the findings.
Generic future. "Future research should…"
Drift or tape. The key noun changes, or a line repeats.

PART III OF III

Link the Review

Where does each later chapter get its material?

Every chapter draws on the review · Orphans and strangers · Pick a Fight

Job Two returns here at full length. With the whole thread in view, the review can be written as its source.

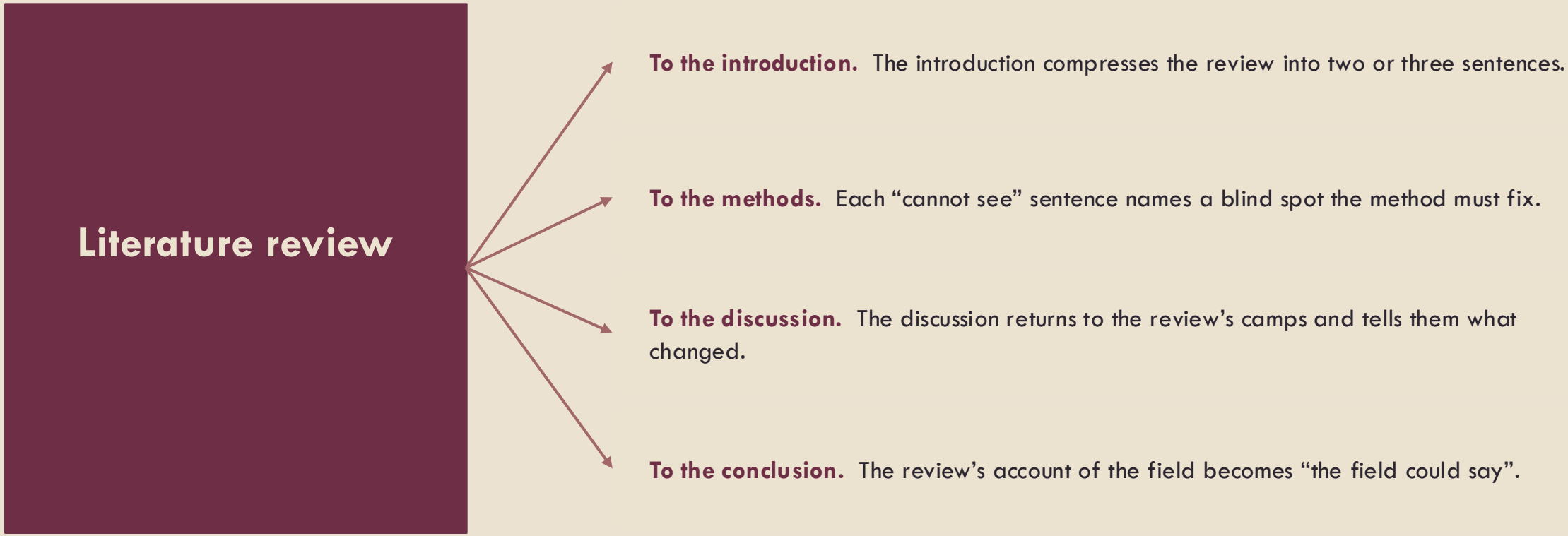
I · EARN THE GAP

II · CARRY THE GAP

III · LINK THE REVIEW

Every Chapter Draws On The Review

The review is not a survey you finish and leave behind. Each later chapter takes something from it.



Write the review as the source the later chapters draw on, not as a survey of the field.

What The Review Says, And Where It Comes Back

The review says	A later chapter answers
"Portfolio research models retirement as an optimisation. Escalation research shows managers failing to perform it."	INTRODUCTION "Innovation research tells firms how to launch. It says almost nothing about how they decide to stop."
"Both literatures infer the decision from which products were still standing afterwards."	METHODS "This study therefore traces each decision through interviews and meeting archives, before the outcome is known."
"Portfolio models predict optimisation. Escalation research predicts resistance."	DISCUSSION "Neither camp predicted what these firms did. They did not optimise and they did not escalate. They deferred."
(The review's closing account of the field.)	CONCLUSION "The field could say that firms keep failing products too long."

The link most often missing is the discussion one. Without it, the discussion talks to new sources and forgets the review's camps.

Orphans & Strangers

Two ways the links between the review and the later chapters break.

The orphan section

A review section that no later chapter uses.

Cut it, or give it a job.

The stranger in the discussion

A source that drives the discussion but never appeared in the review.

Either the review missed it, or the discussion has changed the subject.

Every review section needs a return address, a later chapter that uses it.

Activity Three

Pick a Fight, then send it forward

---

15 minutes. On your own, then in pairs at your table.

Presenting stops here. You are on.

Pick A Fight, Then Send It Forward

Your sheet has a smooth literature review paragraph for your field. Two sources in it disagree, but the paragraph hides the disagreement.

1 **Hunt the collision**

Find the two sources that disagree. State in one sentence what they disagree about.

2 **Make them argue**

Rewrite the paragraph so the sources meet head-on. End with a "therefore" sentence stating the question the fight earns.

3 **Send it to the methods**

One sentence on what kind of study, reading or proof could settle the fight.

4 **Send it to the discussion**

One sentence your discussion would have to say to both camps, whatever you find.

Steps 3 and 4 are return addresses. They show where this review paragraph comes back later in the thesis.

Is The Fight Real?

Read only your three sentences aloud. Your partner reconstructs the fight from them, then checks it.

A real collision

Same question. Same terms. Answers that cannot both hold.

A staged one

Different questions in different settings, dressed up as a contradiction.

Manufactured disagreement is the mirror error of manufactured consensus.

The machine invents agreement. Anxious writers invent fights.

What The Machine Is For

1

Surfacing disagreement

Paste sources you have read. Ask where they conflict. Then check every claimed conflict against the originals.

2

Prosecuting your gap

Run your gap statement against the three tests. It is a better prosecutor than defender.

3

Checking continuity

Paste your gap sentence from each chapter. Ask only whether they make the same claim. Then judge its answer yourself.

What It Is Never For

Finding your gap, or writing the sentences that carry it.

A synthesis tool optimises for plausible consensus, and a gap is an argument against consensus. Ask it for the gap in your field and it will hand you a fluent absence, often one already filled. It also keeps no memory of your gap from one chapter to the next.

Drafting chapter by chapter in separate AI sessions builds drift in by design.

The Session On One Page

Earn the gap

Three tests. Consequence, adjudication, inheritance.
Four cracks. Tension, absence, limit, assumption.
The tool. The crack decides the design, not department habit.

Carry the gap

Key noun. One word, kept in every chapter.
Five jobs. Announce, build, make answerable, rule, account.
Ruling scale. Closed, narrowed, reframed, reopened.
Triad. Could say, can now say, still cannot say.

Link the review

Return address. Every review section is used by a later chapter.
Orphan. A section nothing uses.
Stranger. A discussion source the review never introduced.
Thread, not tape. Carry the claim forward. Do not paste it.

Thank You. Your Questions.

Test the session with its own tests. What did it leave unanswered, and where did it lose its own thread?

Next. Session 3. No preparation needed.

What this session leans on

Alvesson, M. and Sandberg, J. (2011) 'Generating research questions through problematization', *Academy of Management Review*, 36(2), pp. 247–271.

Bunton, D. (2005) 'The structure of PhD conclusion chapters', *Journal of English for Academic Purposes*.

Kwan, B. S. C. (2006) 'The schematic structure of literature reviews in doctoral theses of applied linguistics', *English for Specific Purposes*, 25(1), pp. 30–55.

Lim, J. M. H. (2006) 'Method sections of management research articles: A pedagogically motivated qualitative study', *English for Specific Purposes*, 25(3), pp. 282–309.

Sandberg, J. and Alvesson, M. (2011) 'Ways of constructing research questions: Gap-spotting or problematization?', *Organization*, 18(1), pp. 23–44.

Swales, J. M. (1990) *Genre Analysis: English in Academic and Research Settings*. Cambridge: Cambridge University Press.

The three tests, the four cracks, the five jobs, the key noun, return addresses and the triad are created by the authors'. All examples and exercise passages are teaching fictions. They cite no literature and should not be cited.

The content of this PPTX was human driven. The slides were created and corrected by Grammarly and visualized by Chat GPT

What this session stands on

Alvesson, M. and Sandberg, J. (2011) 'Generating research questions through problematization', *Academy of Management Review*, 36(2), pp. 247–271.

Bunton, D. (2005) 'The structure of PhD conclusion chapters', *Journal of English for Academic Purposes. 4(3), pp. 207–224.*

Kwan, B. S. C. (2006) 'The schematic structure of literature reviews in doctoral theses of applied linguistics', *English for Specific Purposes.*

Lim, J. M. H. (2006) 'Method sections of management research articles. A pedagogically motivated qualitative study', *English for Specific Purposes.*

Sandberg, J. and Alvesson, M. (2011) 'Ways of constructing research questions. Gap-spotting or problematization?', *Organization*, 18(1), pp. 23–44.

Swales, J. M. (1990) *Genre Analysis. English in Academic and Research Settings.* Cambridge University Press.

A deliberately short list. The three tests, the four cracks, the five jobs, the key noun, return addresses and the triad are the authors' frameworks. All examples and exercise passages are teaching fictions. They cite no literature and should not be cited.

The content of this PPTX was human driven. The slides were created and corrected by Grammarly and visualized by Chat GPT